Selling Information in Data Markets

literature survey

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RUC

May 23, 2024

Outline

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1 Motivation

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Data and Information

- The rise of digital platforms such as Amazon, Google in the US and Alibaba, Tencent in China leads to unprecedented collection of individual user data.
- information markets central to economic activity: it takes $20b to acquire and process consumer data (IAB 2018)
- Individual-level data allows companies to refine search results, personalize product recommendations, informative ratings and targeted advertising.

Information Products

- According to a classification first introduced in a Federal Trade Commission report (Fed. Trade Comm. 2014), information products can be distinguished along two key dimensions:

		What does the data broker sell?	
		Only information	**Access to consumer**
Who identifies the prospect?	**Data broker**	Original lists	Sponsored search
	Data buyer	Data appends	Retargeting

Figure 1: Classification of Information Products [1]

- indirect sale: in the form of customized goods and services

[1] reproduced from Dirk Bergemann and Alessandro Bonatti (2019)

Data Markets

There are usually 3 constituent players.

- consumers: have private and possibly imperfect information about their preferences
- firms: offer products to the consumers while choosing prices and quantities
- data intermediaries: collect, aggregate, and distribute information between the consumers and the firms

Research Questions

- several economically relevant questions:
  1. Who buys consumers' information in equilibrium? Does the market enable an efficient use of individual information? (Efficient Use)
  2. How does the social aspect of the data impact the terms of trade between consumers, data buyers, and data intermediaries? (Data Externality)
  3. What determines the value of individual and aggregate data for an information intermediary? (Data Value/Data Pricing)
- Many of these issues have been analyzed separately and in distinct settings, but no unified model.

Motivation

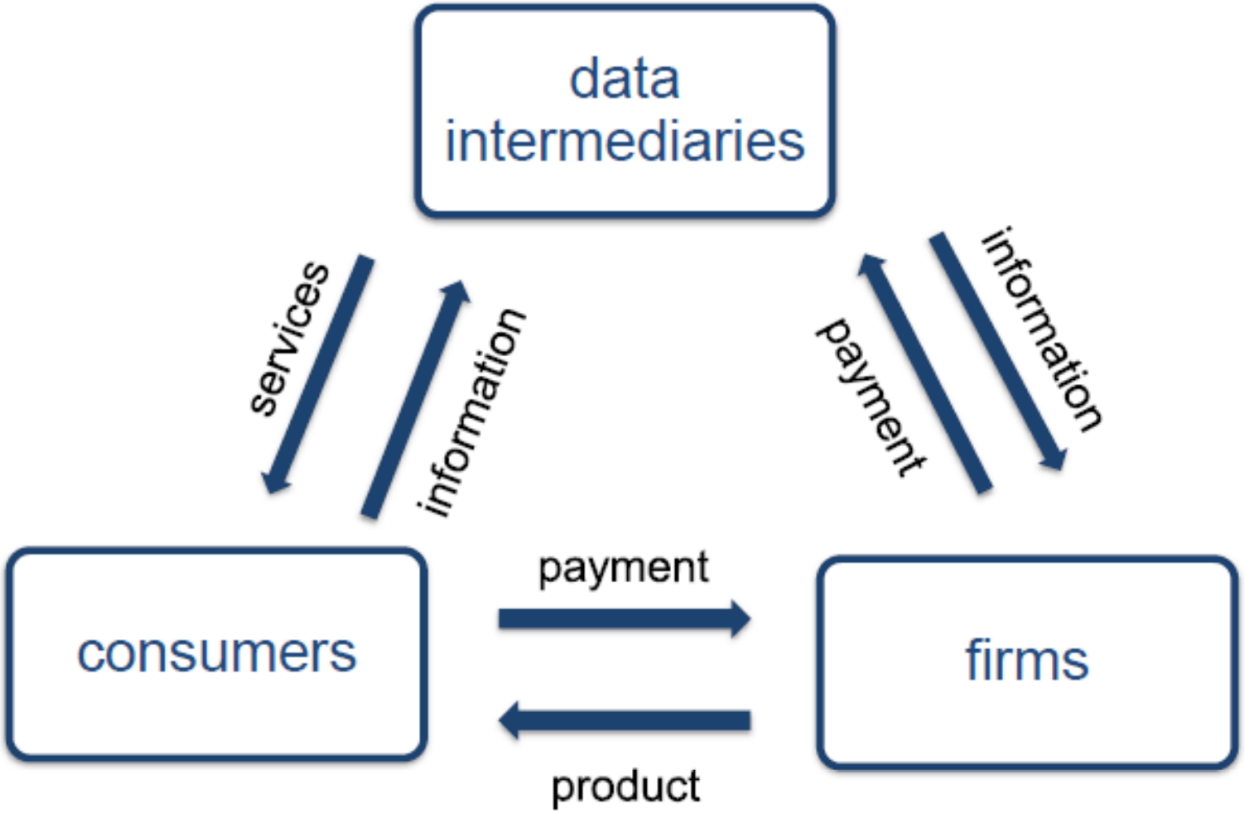

Figure 2: Interaction between the Agents and their Actions

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An Integrated Model

- N consumers, J firms (sellers) and data intermediary
- Each consumer has willingness-to-pay $w_i = \theta + \theta_i$.
- common and idiosyncratic demand shocks θ and θ_i

$$\theta \sim N(\mu, \sigma_\theta^2),\ \theta_i \sim N(0, \sigma_{\theta_i}^2)$$

consumer i chooses quantity q_i to maximize

$$u(w_i, q_i) = w_i q_i - \frac{1}{2} q_i^2 - p_i q_i.$$

- Each firm has a linear production cost c_j and seeks to maximize its expected profit $\pi(q_j, p) = \mathbb{E}[(p - c_j) q_j]$.
- The data intermediary offers a transfer fee $f_i : \mathcal{I}_i \to \mathbb{R}$ to consumer i and asks firm j for a transfer fee $g_j : \mathcal{I}_j \to \mathbb{R}$.

Single Data Intermediary and Single Firm

- With a single firm, the problem essentially becomes one of third-degree price discrimination. Different realizations of the information play the role of market segments.
- The firm, given the estimate about the market demand, forms a linear pricing rule that attempts to extract much of the consumer surplus.
- With the quadratic utility function of the consumers and the constant marginal cost function of the firm, this framework is the classic linear demand problem analyzed by Robinson (1933) and Schmalensee (1981)

Single Data Intermediary and Single Firm

- Robinson (1933) finds that the average quantity supplied is the same with or without price discrimination.
- Schmalensee (1981) finds that in view of lower social welfare due to third-degree price discrimination, there might not be room for a data intermediary to make profits.
- The question is how there can be information sharing and information mediation in equilibrium.
- Bergemann & Bonatti (2018) find that there is always a threshold $\bar{n}$ such that the information intermediary can enter the market and receive positive profits

A Monopolistic (information) Seller and Competing firms

- Admati & Pfleiderer (1986) proposes a model where traders buy information from a monopolistic seller, who is in complete control of the entire database
- The seller may or may not grant access to the information to all buyers. In addition, the seller may or may not add noise to the initial information.
- an oligopoly setting with incomplete information: Bergemann & Morris (2013) analyze the information structure that guarantees the highest industry profit.

A Monopolistic (information) Seller and Competing firms

- Admati & Pfleiderer (1990) extend their analysis to allow for two distinct methods of selling information. They also allow the seller of the information to bundle the information with a product (a portfolio whose composition depends on the available information).
- It also considers the possibility that the seller can use a two-part tariff.
- Admati & Pfleiderer (1988) allow the information seller to either trade their information, sell their information, or both. The seller commits to a policy in advance.

Applications and Variations

- The integrated model allows for many variations, each allowing for a more precise match between the model and the specific information market.
- Sponsored-search auctions: format – Edelman et al. (2007) and Varian (2007)
- Search engines: search term combined with supplemental information about the characteristics of the searching consumer. This increased differentiation leads to the possibility of thinner markets and less competition. Bergemann & Bonatti (2011), Levin & Milgrom (2010), Eliaz & Spiegler (2016)

Discussion

- We restrict attention to multivariate normally distributed variables. The assumption of normality renders the general analysis tractable.
- A heavily parameterized model has relatively large limitations, such as the Gaussian model.
- It reduces the design of the information to a choice of the variance and covariance matrix.

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Mechanism Design

- deal with the choices in the **design of information**
- focus on the interaction between a single information seller and a single buyer
- more general payoffs and more general information structures: non-Gaussian information structures are allowed for

Ex-ante Pricing

- direct sale of information where contracting takes place at the ex ante stage
- In this case, the buyer purchases an information structure instead of paying for specific realizations of the seller's informative signals
- this corresponds to purchasing data appends

Ex-ante Pricing

- Bergemann et al. (2018) consider a model with a single data buyer who can invest in a consumer at fixed conditions.
- Bayesian hypothesis testing: The lender wants to test a null hypothesis H_0 (low-risk borrower) against an alternative H_1 (high-risk).The hypothesis test is characterized by the choice of threshold, which yields type I and type II statistical errors.
- The seller can design any binary (pass/fail) test that reports whether the riskiness measure is above or below a particular threshold.

Ex-ante Pricing

- Bergemann et al. (2018) focus on designing the revenue-maximizing menu for the seller. They also uncover systematic distortions in the information provided under the optimal menu, i.e., in the distribution of states and signals that are associated with monopolistic screening.
- Bergemann & Morris (2019) provide a unified perspective on information design in games.
- A concrete implication: never optimal for the seller to damage information products by adding unbiased noise. Information is degraded by revealing only a portion of the available data to the buyer.

Ex-post Pricing

- The sale of original lists can be modeled as an informative experiment that reveals whether a potential consumer matches a prespecified set of characteristics.
- The buyer receives a contact and pays a price.
- The price paid by the buyer depends on the realization of the seller's information.

Ex-post Pricing

- Bergemann & Bonatti (2015) consider the trade of information bits (browser cookies) that are inputs into a decision problem. A single firm (buyer of information) has heterogeneous match values with a set of consumers.
- Babaioff et al. (2012) study a related model of selling lists (i.e., pricing conditional on signal realizations) when buyers are heterogeneous and privately informed.

Ex-post Pricing

- Eso & Szentes (2007a) and Li & Shi (2017) consider the case where signal realizations are not directly contractible, but the buyer's actions are.
  1. Eso & Szentes (2007a) focus on the case where the seller releases information that is orthogonal to the buyer's type.
  2. Li & Shi (2017) show that discriminatory disclosure of information—providing different buyer types with different signals—dominates full disclosure when the seller is not restricted to orthogonal disclosure.

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Related Qustions

- information sharing between competing firms
- privacy implications of consumer data collection (social nature of data)
- implications for industry structure and for the internal organization of production

Open Qustions

- How does competition among heterogeneous data providers enable firms to better segment their customer populations?
- What are the implications of acquiring an advantage in a downstream market by means of better data?
- Does the ability to access ever more precise predictions and recommendations shrink the boundaries of the firm and enable a platform model?

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Takeaway

- a key difference from normal markets: data intermediary and its market power
- two perspectives to view data markets: Classic Markets and Mechanism Design
- In Classic Markets, we restrict attention to normally distributed variables. The assumption of normality renders the general analysis tractable.
- Mechanism Design zooms in to one of the central elements of information markets: the design of the information.
  1. Ex ante and ex post sale of information are distinguished. The difference between these types relates to the two kinds of products (original lists and data appends).
  2. More general payoffs and more general information structures